MAHATMA GANDHI UNIVERSITY, KOTTAYAM
FIFTH SEMESTER – OCTOBER 2026
MG5DSEFTQ304 – POST HARVEST TECHNOLOGY
ANSWER KEY

Maximum Marks: 70

Part A – Short Answer Questions (2 marks each)

1. Which type of heating occurs during RF processing?

Radio-frequency (RF) processing produces dielectric heating. The alternating electric field causes polar molecules and ions in the food to move or oscillate, generating heat within the material.

2. Define microwave heating.

Microwave heating is a method of dielectric heating in which electromagnetic waves, generally at 915 MHz or 2.45 GHz, interact with polar molecules and ions in food and generate heat throughout the exposed material.

3. Discuss two applications of supercritical fluid extraction in the spice processing industry.

Extraction of essential oils and volatile aroma compounds from spices such as pepper, cardamom and clove.
Extraction of oleoresins, colouring compounds and flavour components from spices such as turmeric, paprika and chilli. Carbon dioxide is commonly used as the supercritical solvent.

4. Define nanosensor.

A nanosensor is a sensing device that uses nanomaterials or nanoscale structures to detect and measure very small amounts of a target substance or change, such as a pathogen, toxin, gas or chemical residue.

5. List out the factors affecting post-harvest loss.

Mechanical injury and improper handling.
Temperature and relative humidity during storage.
Microbial infection and pest attack.
Physiological changes, respiration and transpiration.
Poor packaging, transport and storage facilities.

6. Explain any four areas included in post-harvest management.

Harvesting at the correct maturity stage.
Cleaning, washing and removal of damaged produce.
Sorting and grading according to size, quality and maturity.
Packaging, storage and transportation under suitable conditions.

7. State two factors affecting membrane flux.

Transmembrane pressure difference: increasing pressure generally increases flux up to a limiting level.
Feed characteristics and membrane fouling: higher viscosity, suspended solids and concentration polarization can reduce flux. Temperature and membrane properties also influence flux.

8. Define nanotechnology in the context of the food industry.

Nanotechnology is the application of materials and structures with dimensions typically in the range of 1–100 nm to food processing, packaging, preservation, delivery systems and quality or safety monitoring.

9. Justify the use of plant growth regulators as a method for extending the storage life of fruits and vegetables.

Plant growth regulators can control ripening, senescence and physiological activity. For example, inhibitors of ethylene action or ethylene production can delay ripening and softening, while suitable regulators can reduce sprouting or maintain firmness, thereby extending storage life when correctly applied.

10. What does PEF stand for?

PEF stands for Pulsed Electric Field.

11. Explain any two important properties of supercritical fluid extraction.

Solvent-like density: a supercritical fluid has density comparable to a liquid and can dissolve selected compounds.
Gas-like diffusivity and low viscosity: it penetrates porous materials readily and supports faster mass transfer. Its solvent power can be adjusted by changing pressure and temperature.

12. Recognise one application of nanoparticles in food preservation.

Silver nanoparticles may be incorporated into suitable packaging materials because of their antimicrobial activity, helping to inhibit the growth of spoilage microorganisms. Their use must comply with food-contact safety regulations.

Part B – Short Essay Questions (5 marks each)

13. Explain the importance of proper handling, storage and transportation in maintaining quality.

Proper post-harvest handling, storage and transportation preserve freshness, nutritional value, appearance and safety of produce.
Careful harvesting and handling prevent cuts, bruises and compression damage, which otherwise increase respiration and microbial infection.
Cleaning, sorting and grading remove soil, foreign matter and damaged produce and help maintain uniform quality.
Maintaining the recommended temperature and relative humidity slows respiration, transpiration, ripening and senescence.
Suitable packaging protects produce from physical damage, moisture loss and contamination while allowing appropriate ventilation.
Hygienic transport, reduced delays and proper stacking prevent contamination and mechanical injury.
Thus, good practices reduce quantitative and qualitative losses and improve shelf life and market value.

14. Relate the importance of raw material selection to the quality of processed fruits and vegetables.

The quality of the final processed product depends strongly on the quality and suitability of the raw material.
Variety: select cultivars with suitable flavour, colour, texture, composition and processing characteristics.
Maturity: correct maturity provides appropriate sugar–acid balance, colour, firmness and yield.

Freshness and soundness: fresh, undamaged produce reduces microbial load and prevents off-flavours and defects.
Uniformity: uniform size and maturity support even peeling, cutting, blanching, drying or heating.
Safety: raw materials should be free from pests, pesticide residues, toxins and other contaminants within permitted limits.
Proper selection improves product quality, processing efficiency, yield and shelf life while reducing rejection and waste.

15. Discuss the principle, advantages and applications of Supercritical Fluid Extraction in food processing industry.

Principle: A substance is supercritical when it is above its critical temperature and critical pressure. In this state it combines liquid-like solvent density with gas-like diffusivity. Supercritical carbon dioxide is commonly passed through the raw material to dissolve target compounds; pressure reduction then separates the extract from the solvent.
Advantages: Operates at relatively low temperatures, helping protect heat-sensitive compounds.
Carbon dioxide is non-flammable, relatively non-toxic, readily available and leaves little solvent residue.
Selectivity can be adjusted by changing pressure and temperature. Solvent can be separated and recycled.
Applications:
Decaffeination of coffee and tea.
Extraction of essential oils and oleoresins from spices.
Recovery of flavours, fragrances, antioxidants and other valuable food ingredients.
Removal of unwanted components such as certain lipids from food materials.

16. Explain the significance of nanoparticles in the food processing sector.

Nanoparticles have useful size-dependent properties, including high surface area, reactivity and functional activity.
Food packaging: antimicrobial nanoparticles can help inhibit spoilage organisms.
Barrier improvement: nanofillers can improve packaging strength and reduce oxygen or moisture transmission.
Nutrient delivery: nanoencapsulation can protect sensitive nutrients and improve their stability and controlled release.
Processing: nanoscale materials can support filtration, clarification and delivery of functional ingredients.
Detection: nanosensors can help detect pathogens, toxins, gases and chemical contaminants.
Applications require evaluation of migration, toxicity, exposure and regulatory compliance to ensure consumer safety.

17. Explain the different operations involved in post-harvest management.

Major operations include:
Harvesting: collecting produce at the correct maturity and under suitable conditions.
Pre-cooling: removing field heat to slow respiration and deterioration. Cleaning and washing: removing soil, debris and surface contaminants.
Sorting and grading: separating damaged produce and classifying according to size, maturity and quality.
Treatment: applying suitable curing, disinfection, waxing or other approved treatments where required.
Packaging: using suitable containers to protect produce and facilitate handling. Storage: maintaining appropriate temperature, humidity and atmosphere.
Transportation and marketing: moving produce hygienically with minimal delay and physical damage.

18. Suggest a suitable application of inductive heating in the food industry and explain how the technology can improve processing efficiency.

A suitable application is induction heating of metal food-processing equipment, such as a stainless-steel vessel or a metal surface used for heating a food product. An alternating current in an induction coil creates a changing magnetic field, inducing eddy currents in the conductive metal; electrical resistance converts these currents into heat.

Heat is generated directly in the metal rather than being transferred from a separate flame or hot surface.

Rapid heating and precise temperature control can shorten processing time. Energy losses to the surroundings can be reduced.

The system can provide a clean, flame-free and easily automated heating arrangement.

It may improve process consistency and throughput. The equipment must be designed with food-grade contact surfaces and suitable temperature control.

19. Differentiate between microfiltration and ultrafiltration based on membrane pore size, retained components and applications.

Microfiltration (MF):

Typical pore size: approximately 0.1–10 μm.

Retains suspended particles, larger bacteria and fat globules, depending on the membrane.

Applications include clarification of beverages, removal of bacteria from liquids and milk processing.

Ultrafiltration (UF):

Typical pore size: approximately 0.001–0.1 μm; membranes are often specified by molecular-weight cut-off.

Retains proteins, colloids, macromolecules and many viruses, while water and smaller solutes pass through.

Applications include concentration and fractionation of milk proteins, whey processing and concentration of juices or other food liquids.

MF generally has larger pores and retains larger particles; UF has smaller pores and retains dissolved macromolecules.

20. Apply nanotechnology concepts to develop a suitable approach for food packaging and preservation.

A suitable approach is an active nanocomposite packaging film for fresh or minimally processed food. A biodegradable or food-contact polymer can be reinforced with suitable nanoscale fillers to improve its mechanical and gas/moisture barrier properties. An approved antimicrobial component may be incorporated to reduce microbial growth.

Select a suitable polymer and food-compatible nanofiller.

Disperse the nanofiller uniformly to improve strength and barrier performance.

Incorporate an appropriately assessed antimicrobial agent, if permitted.

Form the film and evaluate tensile strength, permeability, antimicrobial activity and food shelf life.

Conduct migration and toxicological assessments and ensure regulatory compliance.

Expected benefits include reduced contamination, slower quality deterioration, improved package strength and potentially longer shelf life.

Part C – Long Essay Questions (10 marks each)

21. Debate the importance of proper irrigation, fertilization, pest management, and other pre-harvest practices in maintaining post-harvest quality and reducing losses.

Pre-harvest practices influence the physical, chemical and physiological condition of produce at harvest and therefore affect its storage life and market quality.

1. Irrigation: Adequate and timely water supply supports normal growth, size and quality. Water stress may result in small produce, poor development and uneven maturity. Excess irrigation near harvest can cause cracking, low soluble-solids content in some crops and greater disease risk. Irrigation should therefore be scheduled according to crop and soil requirements.
2. Fertilization: Balanced nutrient application supports proper growth, firmness, colour and composition. Excess nitrogen may produce soft tissues and increase susceptibility to damage or diseases, while deficiencies can reduce yield and quality. Potassium, calcium and other nutrients contribute to quality attributes and tissue integrity in many crops.
3. Pest and disease management: Insects, fungi, bacteria and other pests cause wounds, rots and visible defects before harvest. Integrated pest management, field sanitation, monitoring and permitted control measures reduce infestation and the chance of post-harvest decay.
4. Crop and variety selection: Choosing varieties suited to the climate, intended market and storage period improves uniformity, firmness and keeping quality.
5. Orchard/field practices: Pruning, weed control, sanitation, canopy management and appropriate spacing improve light and air movement, reduce disease pressure and support uniform development.
6. Harvest maturity and timing: Harvesting at the correct maturity stage and during suitable weather conditions reduces physiological disorders and mechanical damage and supports the desired ripening and storage behaviour.

Overall, pre-harvest practices cannot eliminate all post-harvest losses, but they establish the initial quality of the produce. Integrated and balanced management improves quality, reduces decay and waste, and increases storage life and market value. Practices must be crop-specific; excessive inputs can have adverse effects.

22. Explain the role of electrical conductivity of food materials in ohmic heating and discuss the factors affecting conductivity.

Ohmic heating is a direct electrical heating method in which an alternating electric current is passed through food placed between electrodes. The food acts as an electrical resistance, and electrical energy is converted into heat within the product (Joule heating). The rate of heat generation per unit volume is related to electrical conductivity and the square of the electric field strength: $q = \sigma E^2$, where q is the volumetric heat-generation rate, σ is electrical conductivity and E is electric-field strength.

Role of electrical conductivity:
Conductivity determines how readily current passes through the food and therefore influences internal heat generation.
Higher conductivity generally produces greater heat generation at a given electric field strength.
Uniform conductivity helps achieve uniform heating; differences between components can cause uneven heating.
Conductivity changes with temperature, so heating behaviour may change during processing.

Factors affecting conductivity:
1. Temperature: conductivity of many foods increases as temperature rises because ionic mobility increases.
2. Ionic content and salt concentration: dissolved salts and mineral ions increase conductivity.
3. Moisture content: water provides a medium for ion movement; low-moisture foods generally have lower conductivity.
4. Composition: the proportions of fat, protein, carbohydrates, acids and minerals affect conductivity.
5. Structure and physical state: particle size, porosity, orientation and contact between particles influence current

pathways.
6. pH and type of ions: ion concentration and mobility affect electrical behaviour.

Ohmic heating can provide rapid and volumetric heating and is useful for liquid foods and particulate foods such as soups, sauces and fruit preparations. Proper control of conductivity, electrode design, voltage and temperature is essential for safe and uniform processing.

23. Evaluate the suitability of reverse osmosis for fruit juice concentration with respect to product quality and processing efficiency.

Reverse osmosis (RO) is a pressure-driven membrane process. Pressure greater than the osmotic pressure is applied to fruit juice so that water passes through a semipermeable membrane while most sugars, acids, flavour compounds and other dissolved solids are retained in the concentrate.

Advantages for product quality:
RO operates at relatively low temperatures, reducing heat damage to colour, flavour, aroma and heat-sensitive nutrients.
It can reduce water content without the extensive thermal exposure associated with evaporation.
It may help retain fresh-like sensory characteristics and nutritional quality.
No phase change is required, which can reduce thermal degradation.

Processing efficiency and limitations:
RO can be energy-efficient compared with conventional thermal concentration, particularly at lower concentration levels.
The process can be operated continuously and may be integrated with other membrane processes.
RO has a practical concentration limit because osmotic pressure and viscosity rise as soluble solids increase.
High viscosity, concentration polarization and membrane fouling reduce permeate flux.
Membrane selection, cleaning, pressure requirements and product pretreatment affect operating cost and performance.
Some aroma compounds or small solutes may pass through the membrane depending on membrane properties and operating conditions.

Evaluation: RO is suitable as a low-temperature pre-concentration step for fruit juice when preservation of fresh flavour and heat-sensitive nutrients is important. It is generally less suitable as the sole method for reaching very high soluble-solids concentrations. A practical process may combine RO with vacuum evaporation or another finishing concentration method. Suitability depends on juice composition, target concentration, membrane performance and economics.

24. Propose a suitable nanocomposite-based packaging application for a selected food product and explain its expected benefits.

Proposed product: fresh strawberries (or another highly perishable fresh fruit).

Packaging design: Develop a food-contact polymer film or tray coating containing a well-dispersed, approved nanoscale inorganic or clay-based filler. The nanofiller can improve mechanical strength and reduce oxygen and water-vapour transmission. An antimicrobial component may be included only if its safety, migration and regulatory status are established. A suitable breathable package design is important because fresh strawberries continue to respire.

Development steps:
1. Select a food-contact polymer suitable for fresh-fruit packaging.
2. Select a compatible nanofiller and disperse it uniformly in the polymer.
3. Prepare the film or coating using an appropriate processing method.
4. Optimize permeability so that oxygen and carbon dioxide levels do not cause undesirable anaerobic conditions.
5. Test mechanical strength, barrier properties, sealability and package performance.
6. Evaluate microbial growth, weight loss, firmness, colour, sensory quality and shelf life during storage.
7. Conduct nanoparticle migration and safety assessments and comply with applicable food-contact regulations.

Expected benefits: Improved strength and durability of the package.
Better control of oxygen and moisture transfer, helping reduce oxidation and moisture loss.
If a safe and permitted antimicrobial function is included, reduced growth of selected spoilage microorganisms.
Better maintenance of freshness, texture and appearance and a possible extension of shelf life.

The packaging must be validated for the particular food. Nanomaterials should not be assumed safe solely because they are effective; migration, exposure and toxicological assessment are essential.